

STATISTICS

Analysis of Variance (ANOVA)

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Rev: 10.12.2024

Confidence Interval

Statistics

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The Means of More Than Two Populations

We apply the t-test (or z-test) to compare the means of **two populations**. When we compare the means of **more than two populations**, one may try one of these applied, multiple times. In this case, however,

- You should apply $C \binom{M}{2}$ individual hypothesis testing.
- The probability of type I error, α , increases as the population number increases.

For example, when $M = 5$ populations and $\alpha = 0.05$,

$$C \binom{5}{2} = 10 \text{ hypothesis testing are applied, and thus confidence level is:}$$
$$(1 - \alpha)^{10} = (0.95)^{10} \cong 0.60 \rightarrow \alpha \cong 0.40$$

Confidence Interval

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Analysis of Variance (ANOVA)

Analysis **of** **va**riance (ANOVA) is a more appropriate method when comparing the means of more than two populations.

Assumptions of ANOVA:

1. The populations have normal distributions.
2. The homogeneity of variances (The populations have equal variances.)
3. The populations are independent. (Samples are also independent and randomly selected.)

Confidence Interval

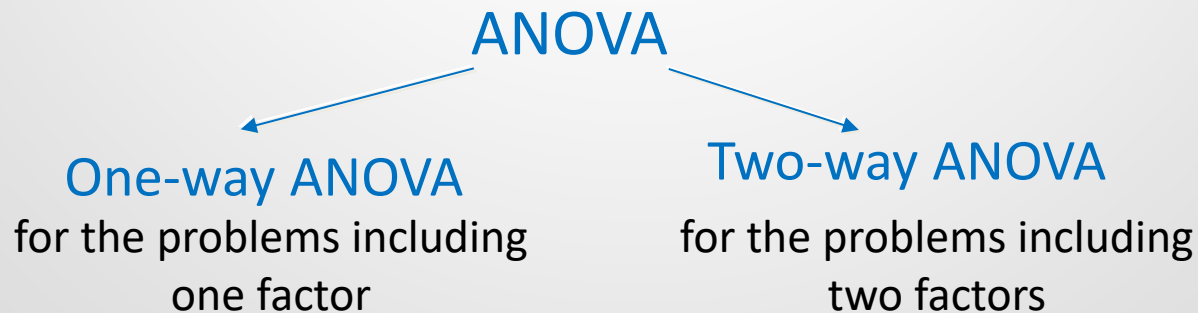
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Types of ANOVA

Consider the following examples to which ANOVA can be applied:

- The expected life of tires from 5 different brands: **One factor** (the brands) affects the results.
- The mean grades in Statistics, Mathematics and English: **One factor** (the type of course) affects the results.
- The mean grades in Statistics, Mathematics, and English for male and female students: **Two factors** (the type of course and the gender) affect the results.
- The relationship between health insurance payments of a person with the career type and marital status: **Two factors** (the career type and marital status) affect the health insurance payments.



Analysis of Variance (ANOVA)

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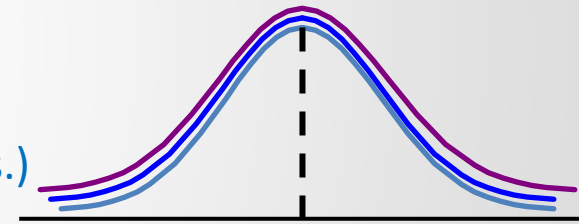
One-Way ANOVA

Consider M populations having normal distribution with equal variances and means $\mu_1, \mu_2, \mu_3, \dots, \mu_M$.

Hypotheses:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_M$$

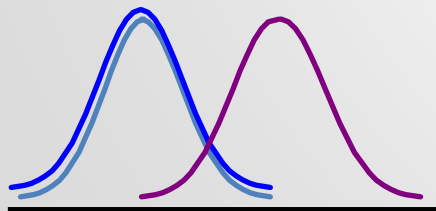
(The factor has no effect on the means of the populations.)



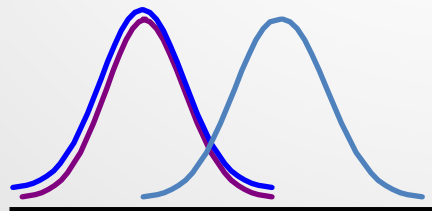
$$\mu_1 = \mu_2 = \mu_3$$

H_a : At least one of the population mean is different

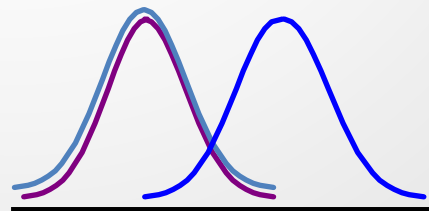
(The factor affects the means of the populations.)



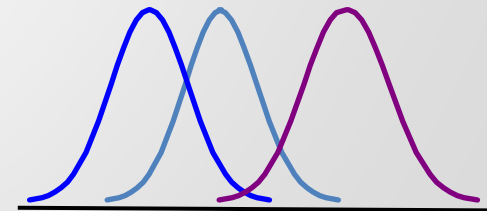
$$\mu_1 = \mu_2 \neq \mu_3$$



$$\mu_1 = \mu_3 \neq \mu_2$$



$$\mu_2 = \mu_3 \neq \mu_1$$

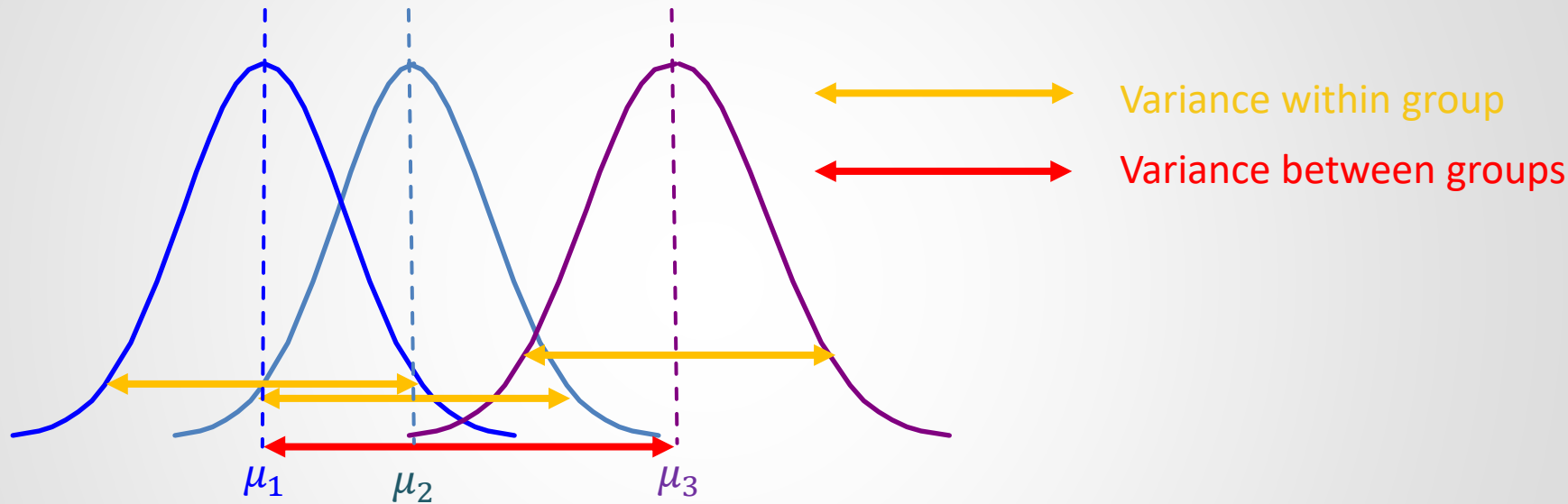


$$\mu_1 \neq \mu_2 \neq \mu_3$$

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Test Statistic:

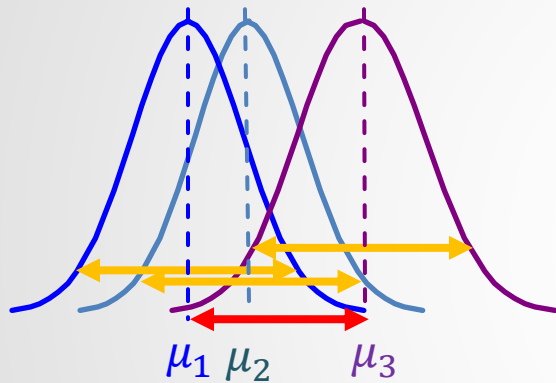
$$F_0 = \frac{\text{Variance Between Groups}}{\text{Variance Within Group}}$$

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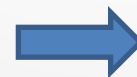
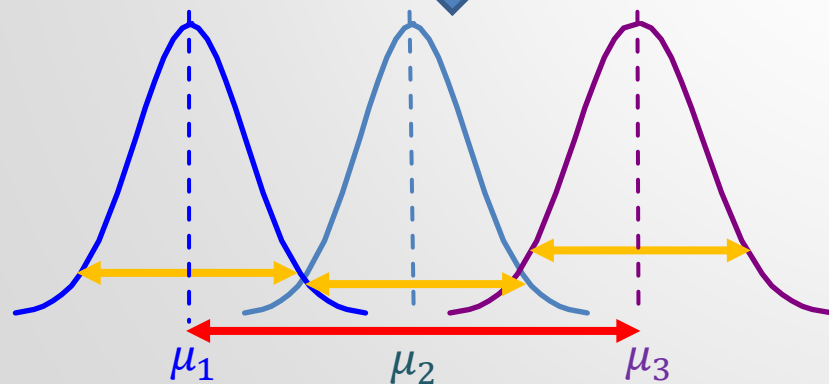
As variance between groups increases, test statistic F_0 increases.



$$F_0 = \frac{\text{Variance Between Groups}}{\text{Variance Within Group}}$$



The means diverge from each other.



implies that H_a is correct

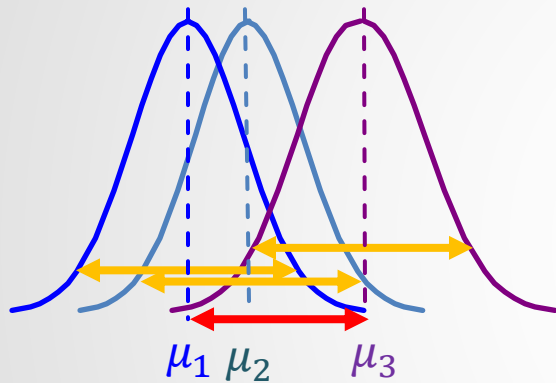
H_a : At least one of the population mean is different
(The factor affects the means of the populations.)

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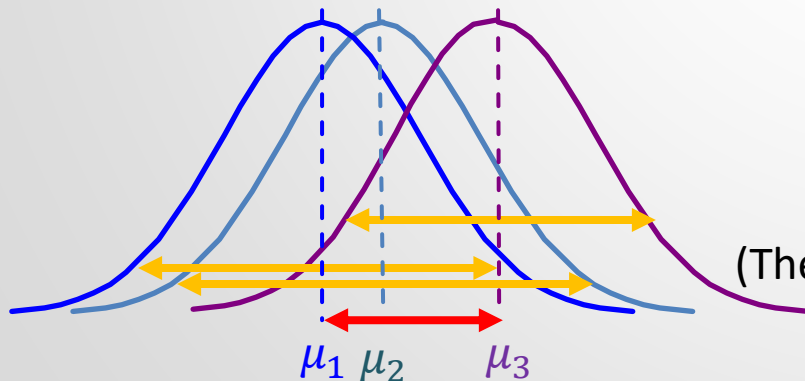
As variance within group increases, test statistic F_0 decreases.



$$F_0 = \frac{\text{Variance Between Groups}}{\text{Variance Within Group}}$$



The intersection area of the populations is increased.



implies that H_0 is correct

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_M$$

(The factor has no effect on the means of the populations.)

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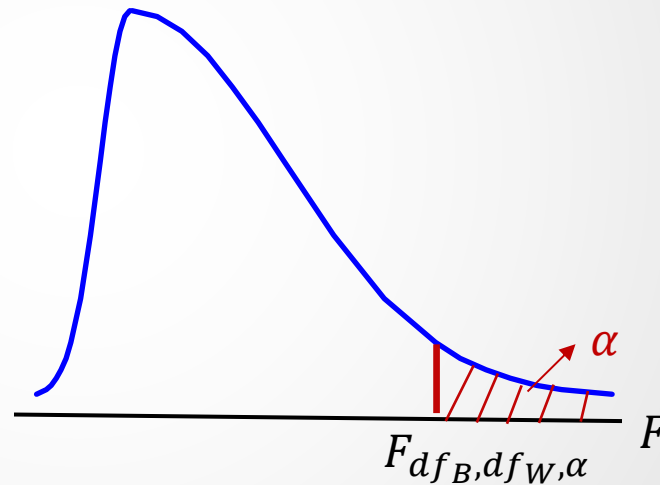
Decision Rule:

If

$$F_0 > F_{df_B, df_W, \alpha}$$

reject H_0 .

Otherwise, fail to reject H_0 .



$$P(F_0 \leq F_{df_B, df_W, \alpha}) = 1 - \alpha$$

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Computation of F_0

The measurements of k samples are given below.

Factor 1					
	Group 1	Group 2	...	Group k	
	x_{11}	x_{21}	...	x_{k1}	
	x_{12}	x_{22}	...	x_{k2}	
	...	...	...	...	
	...	x_{2n_2}	...	...	
	x_{1n_1}		...	x_{kn_k}	TOTAL
sample size	n_1	n_2	...	n_k	$n = \sum_{i=1}^k n_i$
sum of sample	$T_1 = \sum_{j=1}^{n_1} x_{1j}$	$T_2 = \sum_{j=1}^{n_2} x_{2j}$	...	$T_k = \sum_{j=1}^{n_k} x_{kj}$	$T = \sum_{i=1}^k T_i$
sample mean	$\bar{x}_1 = T_1/n_1$	$\bar{x}_2 = T_2/n_2$	...	$\bar{x}_k = T_k/n_k$	$\bar{x} = T/n$

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Computation of F_0

$$F_0 = \frac{\text{Variance Between Groups}}{\text{Variance Within Group}} = \frac{SS_B / df_B}{SS_W / df_W}$$

The sum of squares between groups:

$$SS_B = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2 = n_1 (\bar{x}_1 - \bar{x})^2 + n_2 (\bar{x}_2 - \bar{x})^2 + \cdots + n_k (\bar{x}_k - \bar{x})^2$$

The degree of freedom between groups:

$$df_B = k - 1$$

The sum of squares within group:

$$SS_W = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 = \sum_{i=1}^k (n_i - 1) s_i^2 \text{ where } s_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

The degree of freedom within groups:

$$df_W = n - k$$

the variance of the sample (group) i

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Computation of F_0

The total sum of squares:

$$SS_T = SS_B + SS_W$$
$$\sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2 = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

The total degree of freedom:

$$df_T = df_B + df_W$$
$$n - 1 = k - 1 + n - k$$

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Proof for $SS_T = SS_B + SS_W$

$$\sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2 = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

$$\begin{aligned} SS_B + SS_W &= \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 \\ &= \cancel{\sum_{i=1}^k n_i (\bar{x}_i)^2} - 2\bar{x} \sum_{i=1}^k n_i \bar{x}_i + (\bar{x})^2 \sum_{i=1}^k n_i + \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij})^2 - 2 \sum_{i=1}^k \bar{x}_i \sum_{j=1}^{n_i} x_{ij} + \sum_{i=1}^k (\bar{x}_i)^2 \sum_{j=1}^{n_i} 1 \\ &\quad \bar{x}_i = \frac{\sum_{j=1}^{n_i} x_{ij}}{n_i} \quad n_i = \sum_{j=1}^{n_i} 1 \\ &= -2\bar{x} \sum_{i=1}^k \sum_{j=1}^{n_i} x_{ij} + (\bar{x})^2 \sum_{i=1}^k \sum_{j=1}^{n_i} 1 + \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij})^2 \\ &= \sum_{i=1}^k \sum_{j=1}^{n_i} [(x_{ij})^2 - 2\bar{x}x_{ij} + (\bar{x})^2] = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2 = SS_T \end{aligned}$$

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Mean Squares (variance)

The mean squares between groups:

$$MS_B = \frac{SS_B}{df_B} = \frac{SS_B}{k - 1}$$

The mean squares within groups:

$$MS_W = \frac{SS_W}{df_W} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \cdots + (n_k - 1)s_k^2}{(n_1 - 1) + (n_2 - 1) + \cdots + (n_k - 1)} = \frac{SS_W}{n - k}$$

the weighted mean of the variances s_i^2

$$F_{df_B, df_W, \alpha} = F_{k-1, n-k, \alpha}$$

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One-Way ANOVA Table

Source of the variability	Sum of Squares (SS)	Degrees of freedom (df)	Mean Square (Variance)	F_0
Between Groups	SS_B	$k - 1$	$MS_B = \frac{SS_B}{k - 1}$	$\frac{MS_B}{MS_W}$
Within Groups (Residual-Error)	SS_W	$n - k$	$MS_W = \frac{SS_W}{n - k}$	
Total	SS_T	$n - 1$		

k : the number of populations (groups)

n : the sum of sample sizes

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The Total Variation

Total variance = Variance Between Groups + Variance Within Group

↙
The variation due to the effect
of the factor
+

The variation due to the
random errors (chance factor)

↘
The variation due to the
random errors (chance factor)

When H_0 is correct,

$$F_0 = \frac{\text{Variance Between Groups}}{\text{Variance Within Group}} \approx 1$$

$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_M$
(The factor has no effect on the
means of the populations.)

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The Mathematical Model of One-Way ANOVA

The problem addressed in ANOVA can be modeled by the following formula:

$$X_{ij} = \mu_i + \varepsilon_{ij}$$

where X_{ij} is the j .member of sample i , μ_i is the mean of the population i and ε_{ij} is the random error in the j .member of sample i . Then, if the mean of all populations μ (when the factor has no effect) is subtracted from both sides,

$$X_{ij} - \mu = \mu_i + \varepsilon_{ij} - \mu$$

$$X_{ij} = \mu + (\mu_i - \mu) + \varepsilon_{ij}$$

If we substitute $\alpha_i = \mu_i - \mu$, then

$$X_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

Here, α_i is the effect of the factor for the group i . Thus, hypotheses can be expressed as:

$$H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_M = 0$$

$$H_a: \text{at least one of } \alpha_i \neq 0$$

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ANOVA vs MANOVA

The mathematical model of ANOVA is:

$$X_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

This formula contains a single dependent variable, X_{ij} .

When the formula contains a multivariate dependent variable, **multivariate analysis of variance (MANOVA)** should be used*.

(*) **MANOVA** is not handled in this course.

Hypothesis Testing

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Example 1: The out-of-service times of 23 photocopiers of three different brands in a company in the last year were determined in minutes, and the following data set was obtained. Can we say that there is a difference between the three machines in terms of the mean out-of-service times? Also, test the hypothesis by computing the p-value.

Machine A	Machine B	Machine C
64	45	75
72	62	85
81	73	83
92	58	78
80	50	96
73	68	88
	48	83
	52	84
	66	

(Assume that populations have normal distributions with equal variances and $\alpha = 0.01$)

Hypothesis Testing

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Machine A	Machine B	Machine C	
64	45	75	
72	62	85	
81	73	83	
92	58	78	
80	50	96	
73	68	88	
	48	83	
	52	84	
	66		
$n_1 = 6$	$n_2 = 9$	$n_3 = 8$	$n = 23$
$T_1 = 462$	$T_2 = 522$	$T_3 = 672$	$T = 1656$
$\bar{x}_1 = 77$	$\bar{x}_2 = 58$	$\bar{x}_3 = 84$	$\bar{x} = 72$

$H_0: \mu_1 = \mu_2 = \mu_3$ (The machine brands have no effect on the out-of-service times.)

$H_a: \mu_i \neq \mu_j$ for any $i \neq j$
(The machine brands affect the out-of-service times.)

$$\bar{x}_1 = T_1/n_1 = 462/6 = 77$$

$$\bar{x}_2 = T_2/n_2 = 522/9 = 58$$

$$\bar{x}_3 = T_3/n_3 = 672/8 = 84$$

$$\bar{x} = T/n = 1656/23 = 72$$

$$\begin{aligned}
 SS_B &= \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2 \\
 &= 6(77 - 72)^2 + 9(58 - 72)^2 + 8(84 - 72)^2 \\
 &= 3066
 \end{aligned}$$

$$df_B = M - 1 = 3 - 1 = 2$$

Hypothesis Testing

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Machine A	Machine B	Machine C	
64	45	75	
72	62	85	
81	73	83	
92	58	78	
80	50	96	
73	68	88	
	48	83	
	52	84	
	66		
$n_1 = 6$	$n_2 = 9$	$n_3 = 8$	$n = 23$
$T_1 = 462$	$T_2 = 522$	$T_3 = 672$	$T = 1656$
$\bar{x}_1 = 77$	$\bar{x}_2 = 58$	$\bar{x}_3 = 84$	$\bar{x} = 72$

$$\begin{aligned}
 SS_W &= \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 \\
 &= (64 - 77)^2 + (72 - 77)^2 + (81 - 77)^2 \\
 &\quad + (92 - 77)^2 + (80 - 77)^2 + (73 - 77)^2 \\
 &\quad + (45 - 58)^2 + (62 - 58)^2 + (73 - 58)^2 \\
 &\quad + (58 - 58)^2 + (50 - 58)^2 + (68 - 58)^2 \\
 &\quad + (48 - 58)^2 + (52 - 58)^2 + (66 - 58)^2 \\
 &\quad + (75 - 84)^2 + (85 - 84)^2 + (83 - 84)^2 \\
 &\quad + (78 - 84)^2 + (96 - 84)^2 + (88 - 84)^2 \\
 &\quad + (83 - 84)^2 + (84 - 84)^2 = 1514
 \end{aligned}$$

$$df_W = n - M = 23 - 3 = 20$$

Analysis of Variance (ANOVA)

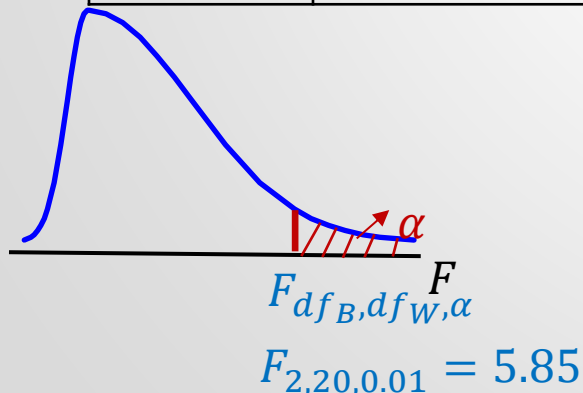
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Example 1:

One-Way ANOVA Table

	Sum of Squares (SS)	Degrees of freedom (df)	Mean Square (Variance)	F_0
Between Groups	3066	$3 - 1 = 2$	$\frac{3066}{2} = 1533$	$F_0 = \frac{1533}{75.7} \cong 20.25$
Within Groups	1514	$23 - 3 = 20$	$\frac{1514}{20} = 75.7$	
Total	4580	$23 - 1 = 22$		



Since $F_0 = 20.25 > F_{df_B, df_W, \alpha} = 5.85$,
we reject H_0 .

We have very significant proof that there is a difference between the three machines in terms of the mean out-of-service times.

Analysis of Variance (ANOVA)

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Example 1:

F Table

$$F_{2,20,0.01} = 5.85$$

$$F_{2,20,0.005} = 6.99$$

$$F_{2,20,p \text{ value}} = F_0 = 20.25 \quad \Rightarrow \quad p \text{ value} \approx 0$$

Since $p \text{ value} < \alpha$, we reject H_0 .

We have a very significant proof that there is a difference between the three machines in terms of the mean out-of-service times.